

Bursting and budding pathogenesis

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Contents

0.1	Gompertz Poisson first jump and expected value	5
1	Replicator suppressor	8
1.1	Model definition	8
1.2	Mathematical definition	9
2	Fixed removal	15
2.1	Successive birth death catastrophe	15
2.2	Extracellular removal	16
3	Note about quad tree sampling	23

The basic model of viral replication was developed in the early 1990s during research into the functioning of HIV within the body. It assumes a constant release of infectious particles from infected cells. The model is deterministic, contained in a set of three ODEs, but is consistent with an underlying stochastic model as the expected process behaviour. In this underlying model, each infected cell releases virions according to a poisson process.

Despite being quite effective, this assumption is only an approximation of reality and is non-cognisant of what happens within cells. It's wide spread use in the BMVR may have even wrongly painted our understanding of how viral release really takes place. In reality, the rate of viral release will not be at its full capacity at the moment of inoculation. Some models have addressed this by incorporating an "eclipse phase" during which the cell is non-productive. It also seems possible that the intracellular load of an infected cell could have something to do with its production rate. This will of course differ between viruses and intracellular bacterial infections, and even between species of each.

A barrier to understanding viral release in particular is that viruses are incredibly small and are not easily observed, as is generally (just about) possible for bacteria. Despite this difficulty, a group at Los Alamos has recently published work attempting to quantify exactly this: the number of HIV particles produced by single infected cells over time. What they found was that the release behaviour is not at all constant, but stochastic and intermittent. Large packets of varying size are released in apparent bursts, at times intervened by relative inactivity.

This chapter explores the fundamental differences between two extremes of the virality strategy space: constant "budding" release, and "all at once" burst release. And a couple of mathematically framed hypotheses are presented which explain potential advantages to so called bursting and intermittent virality.

- **Other work and lack of "additional effects"** Comparing bursting and budding virality is not our idea. There is a niche of preexisting literature which does include stochastic models of each regime and the relative differences seen in analysis. This section overviews some of the work that has previously been done in the area, and identifies only a limited consideration for "additional effects". That is, factors significant to overall viral success that go beyond merely the microscopic replication rate within cells, and the release behaviour. Namely, depletable toxic granules within cells, and locally finite immune factors in the extracellular medium. My work over the following sections proposes additional mathematical quantifications to what has been termed by "(insert name)" as the advantage of "flooding". Or how in a burst style release, there is higher likelihood of using up the local capacity of the host's immune system. [1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] [13] [14] [15] [16] [17] [18] [19] [20] [21] [22] [23] [24] [25] [26] [27] [19] [28] [29] [30] [31] [32].

- **Replicator suppressor**

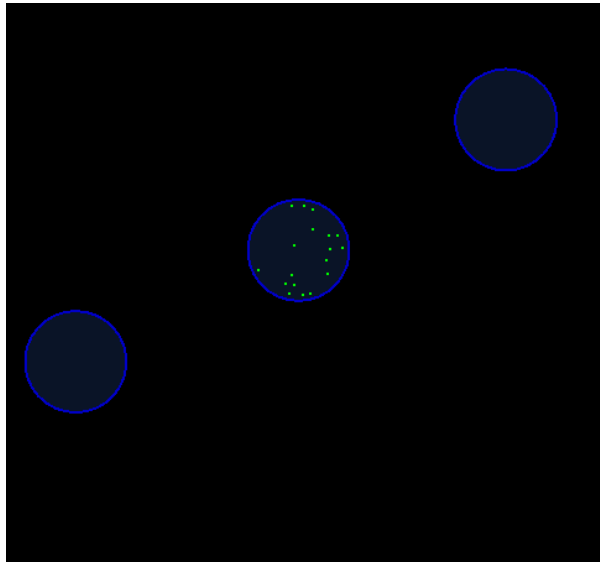
Over the handful of leucocyte types known to science, most contain packets of toxic material called granules (of course, not least, granulocytes). These generally oxidative substances are employed by the immune cells to attack invading pathogens. The leucocytes of interest in our research, macrophages and neutrophils, both contain granules. Neutrophils contain more and use them slightly differently, lysing themselves to release the inflammatory packets to the extracellular medium. Whereas macrophages employ the toxic granules intracellularly, merging the bags of toxin with phagosomal membranes which harbour pathogen after engulfing has taken place. I believe Neutrophils do this as well in addition to the lyse method. And it is this dynamic which forms the basis of the model presented here.

Basically, picture a leukocyte which has just engulfed some number of pathogen. It also contains a count of toxin containing granules which are assumed to be depleted in the process of being used to kill pathogen. With the model specification as given, it can be shown that the number of incident pathogen has a substantial effect of the survival potential of the infection.

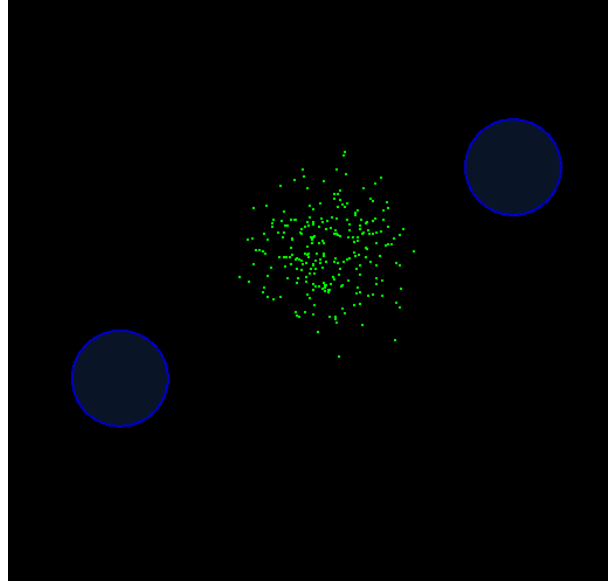
- **N = 2 results**

We were interested in describing an infection across multiple containing cells. For this aim, we began to consider the simple case of an infection going from one cell to another. A birth death catastrophe process in the first which once completed leads to the instantiation of a second BDC process. It's as if all of the bacteria were to go from one cell to the next immediately. There are four quantities of this model: the burst time and size in both the first and second process; t_1 , r_1 & t_2 , r_2 . The first two of these can be found in distribution using content of chapter 2. The second burst quantities are more challenging, but we have made progress in finding their distributions.

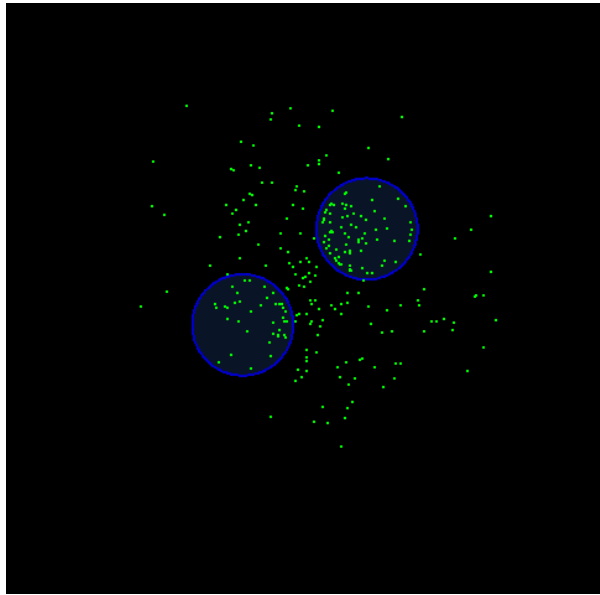
Though this part of the work didn't lead to a grand new model for describing multi-intracellular infection dynamics, it did in part inspire the content of the next section, which takes the concept of successive rounds of a BDC process and incorporates an additional effect.



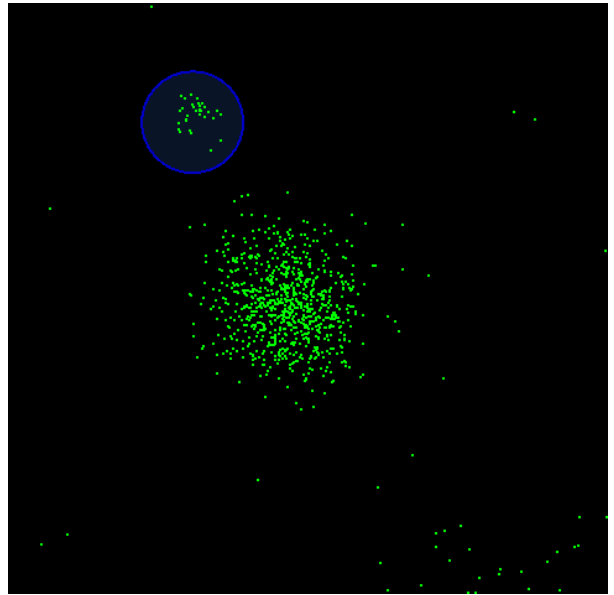
(a)



(b)



(c)



(d)

Figure 1: (a), Intracellular replication before bursting. (b) Cell bursts, and the contents spans the extracellular space before finding new hosts. (c) Macrophages are motile. They wander around the extracellular matrix, phagocytosing debris and extracellular pathogens alike. (d) A second round of cell bursting is displayed, in this case, the second load is larger.

Three non-basic modifications

Here we shall seek to describe rigorously the non-extinction probabilities though time of processes with varied initial state that are subject to non-linear dynamics. Basically, a number of replicative units found an infection within a host cell, their dynamics accord with a continuous-time birth death process. What is the probability at time t there is at least one unit still alive, i.e. “the infection” has

not died out? This will depend on the criticality of the process — whether a given unit is expected to leave behind greater or less than a replacement of itself — which will depend on the birth and death rates, λ and μ . But we are here interested in the effect of process non-linearities, and what implications these have for survivability at varied initial counts, $\mathbf{X}_0 = n$.

Taking the standard birth-death process, $\mathbf{X}_t$, for now with one initial replicative unit, $\mathbf{X}_0$. Its transition rates are defined

$$\begin{aligned}\Pr\{\Delta\mathbf{X}_t = 1\} &= \lambda\mathbf{X}_t\Delta t + o(\Delta t), \\ \Pr\{\Delta\mathbf{X}_t = -1\} &= \mu\mathbf{X}_t\Delta t + o(\Delta t), \\ \Pr\{\Delta\mathbf{X}_t = 0\} &= 1 - (\lambda + \mu)\mathbf{X}_t\Delta t + o(\Delta t),\end{aligned}$$

(with $\Delta\mathbf{X}_t = \mathbf{X}_{t+\Delta t} - \mathbf{X}_t$). In this linear-format, the process is said to be critical exactly when $\lambda = \mu$. Each replicative unit eventually dies or divides, so when chances for each of these eventualities balances, the expected number of “offspring” for a given unit will be 1. More rigorously, defining the count of progeny units left behind as Z :

$$\begin{aligned}\Pr\{Z = 2\} &= \frac{\lambda}{\lambda + \mu} \\ \Pr\{Z = 0\} &= \frac{\mu}{\lambda + \mu}\end{aligned}$$

Left behind should be read as “following the next event”, so time become irrelevant. When $\lambda = \mu$,

$$\Pr\{Z = 2\} = \Pr\{Z = 0\} = \frac{\lambda}{2\lambda} = \frac{1}{2}.$$

And the expected replacement count for a particles is 1;

$$\mathbb{E}(Z) = 0 \cdot \Pr\{Z = 0\} + 2 \cdot \Pr\{Z = 2\} = 2 \cdot \frac{1}{2} = 1.$$

If $\lambda > \mu$, this value is greater than 1, and if $\lambda < \mu$, it’s less than 1, these cases are termed supercritical and subcritical respectively.

$\mathbb{E}(Z) > 1$: **Supercritical.** Given on-average replacement and increase, a population may avoid extinction indefinitely. This occurring with some probability $S(t)$ as defined below.

$\mathbb{E}(Z) = 1$: **Critical.** Despite each unit fully replacing itself in expectation, a critical birth death chain will inevitably decline to nothing. Interestingly, the expected time until this happens is infinite.

$\mathbb{E}(Z) < 1$: **Subcritical.** On average, each unit will not replace itself and the process will decline, though stochastically, with ultimate demise — fixation at 0.

0.1 Gompertz Poisson first jump and expected value

The Poisson process, as $\mathbf{X}_t$ is defined

$$\begin{aligned}\Pr\{\Delta\mathbf{X}_t = 1\} &= \lambda\Delta t + o(\Delta t), \\ \Pr\{\Delta\mathbf{X}_t = 0\} &= 1 - \lambda\Delta t + o(\Delta t);\end{aligned}$$

a constant rate of stochastic incidence events.

0.1.1 First jump probability

Defining $\Pr\{\mathbf{X}_t = i\} = p_i(t)$, the move from state 0 to 1 can be quantified with the state zero probability, $p_0(t)$. And this can be computed in the following way.

$$p_0(t + \Delta t) = p_0(t)[1 - \lambda\Delta t + o(\Delta t)].$$

In plain English: the probability $\mathbf{X}_{t+\Delta t} = 0$ is the likelihood $\mathbf{X}_t = 0$, and, in the interval $(t, t + \Delta t]$, nothing happens. There is no “immigration” event. When taken in the limit $\Delta t \rightarrow 0$, this formula derives the ODE,

$$\begin{aligned} \frac{p_0(t + \Delta t) - p_0(t)}{\Delta t} &= \lambda p_0(t) + o(\Delta t) \\ \xrightarrow{\Delta t \rightarrow 0} \frac{dp_0(t)}{dt} &= \lambda p_0(t). \end{aligned}$$

This is simple, of course. And the solution is a basic decaying exponential, $p_0(t) = e^{-\lambda t}$. But what would happen if we modified the addition rate, making the process inhomogeneous in time. Say for example the rate itself was a decaying exponential: $\lambda \rightarrow \lambda(t) = e^{-t}$?

$$\begin{aligned} \Pr\{\Delta \mathbf{X}_t = 1\} &= \lambda(t)\Delta t + o(\Delta t), \\ \Pr\{\Delta \mathbf{X}_t = 0\} &= 1 - \lambda(t)\Delta t + o(\Delta t); \end{aligned}$$

The question is: can we make the same simple derivation of $p_0(t)$ without issue? It’s not obvious at first that everything will work out. When writing the first step equation,

$$p_0(t + \Delta t) = p_0(t)[1 - \lambda(?)\Delta t + o(\Delta t)],$$

there is some ambiguity about what should be in the argument of $\lambda(t)$. Do we take the value at the start, or the end of the interval: $\lambda(t)$ or $\lambda(t + \Delta t)$? The probability that an event happens or not during the interval can not simply be defined by either of these. But in all likelihood, we are confusing ourselves for no reason. Because the equation is ultimately evaluated in the limit, it shouldn’t matter. $\lim_{\Delta t \rightarrow 0}(t + \Delta t) = t$, so the two cases end up being identical. It seems obvious when put like this, I just hesitate because things aren’t always as they seem in probability theory and we wouldn’t want to be caught out. In any case, it is possible and straightforward to verify the subsequent result.

With the inhomogeneous version of the equation,

$$\frac{dp_0(t)}{dt} = \lambda(t)p_0(t), \quad \lambda(t) = e^{-\alpha t},$$

$p_0(t)$ can be computed as

$$p_0(t) = e^{(e^{-t}-1)}$$

With $\lambda(t) = \beta e^{-\alpha t}$, this generalises to

$$p_0(t) = e^{\frac{\beta}{\alpha}(e^{-\alpha t}-1)}.$$

Poisson Gompertz

$$\begin{aligned} \Pr\{\Delta \mathbf{X}_t = 1\} &= \lambda(t)\Delta t + o(\Delta t), \\ \Pr\{\Delta \mathbf{X}_t = 0\} &= 1 - \lambda(t)\Delta t + o(\Delta t); \end{aligned}$$

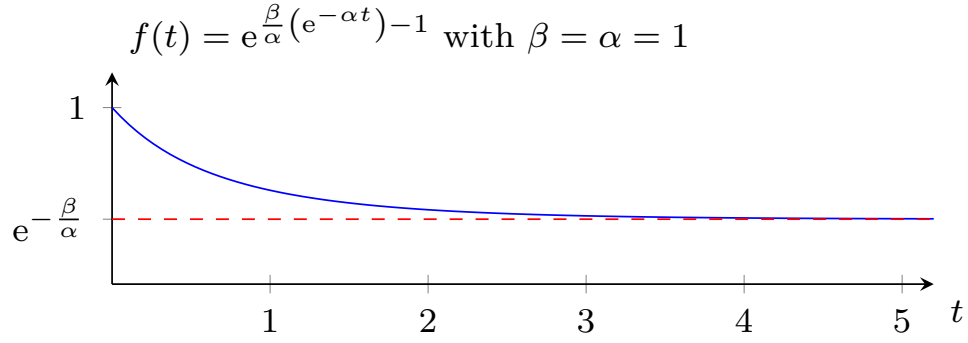


Figure 2: Decaying exponential function $f(t) = e^{e^{-t}-1}$.

No added constant

$$\lambda(t) = \beta e^{-\alpha t}$$

$$\mathbb{E}(\mathbf{X}_t) = \frac{\beta}{\alpha} (1 - e^{-\alpha t})$$

$$p_0(t) = e^{\frac{\beta}{\alpha}(e^{-\alpha t} - 1)}.$$

Added constant

$$\lambda(t) = \beta e^{-\alpha t} + \gamma$$

$$\mathbb{E}(\mathbf{X}_t) = \frac{\beta}{\alpha} (1 - e^{-\alpha t}) + \gamma t$$

Deterministic version

Considering a single unit exhibiting Gompertz activation.

$N(t)$: activation of a node.

$$\frac{dN(t)}{dt} = (\lambda(t) - \mu) N(t)$$

- Gompertz growth rate: $\lambda(t) = \beta e^{-\alpha t} + \gamma$. Made up of,
 - $\beta e^{-\alpha t}$: decaying term represents the reduced proliferation in response to recognition and desired ceasing, and
 - γ : constant, a sort of base-line growth rate.
- μ : decline rate.

Formula:

$$N(t) = N_0 e^{\left[\frac{\beta}{\alpha}(1 - e^{-\alpha t}) + (\gamma - \mu)t\right]}$$

Interesting case: When $\mu > \gamma$ and $\mu < \gamma + \beta e^{-\alpha t}$ when $t = 0$. Sufficient to say $\gamma < \mu < \gamma + \beta$.

1 Replicator suppressor

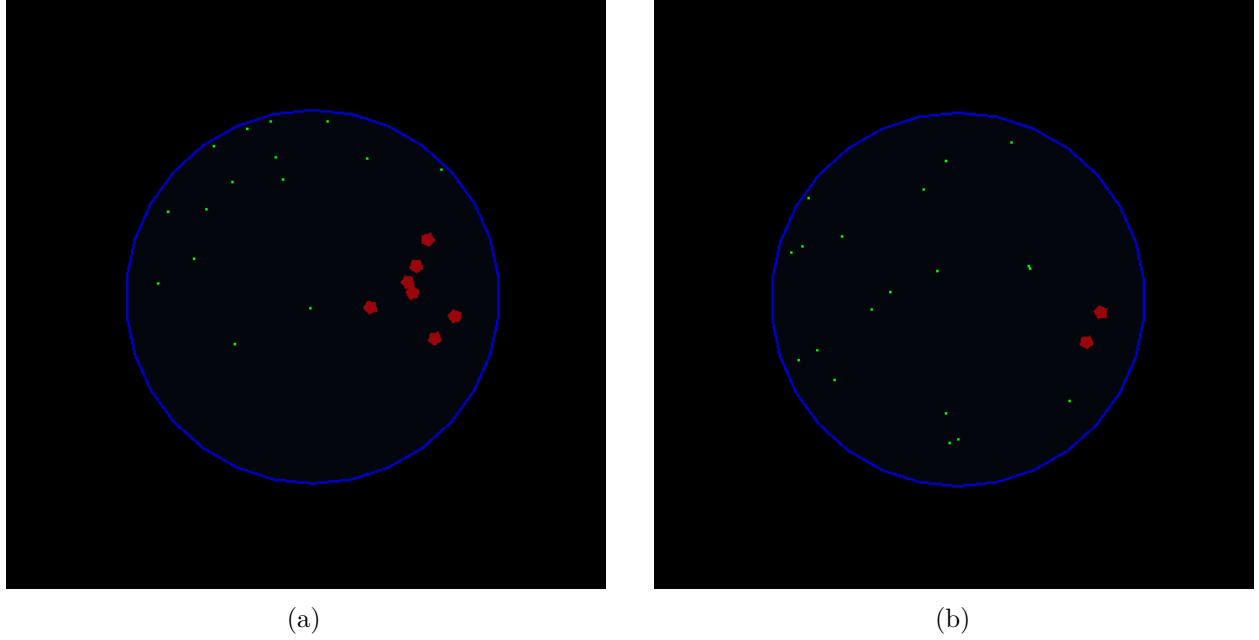


Figure 3: (a), Intracellular granules are represented in red, pathogenic particles in green. In this simulation, when one of each contacts, they anniahilate. (b), the pathogen has replicated and appears to be overcoming the immune challenge of the granules.

1.1 Model definition

Both macrophages and neutrophils contain small a number of small packets called granules. These contain oxidative chemicals and are used to kill engulfed pathogen. Neutrophils are of the category “granulocyte” and as the name suggests contain a larger number of these packets than macrophages.

When a count of pathogenic units enters a macrophage by phagocytosis, they begin to face the the action of these granules. Given the packets are finite in number and depletable, it might be that a larger incident cohort of replicators has a greater chance of eventual survival.

Here we explore a simple model which represents the interplay between the two opposing forces. Replicative pathogenic particles fighting for survival against its host cell’s inbuilt defence mechanism. The model thus constitutes two unit types: $\mathbf{X}_t$ represents the infectious agents, $\mathbf{Y}_t$ refers to the remaining number of within-cell granules. From a defined initial condition $(\mathbf{X}_0, \mathbf{Y}_0)$, the dynamic between these two factors results in either pathogen extinction, or the overthrowing of immunity resulting in unbounded growth in $\mathbf{X}_t$. Of course, in reality growth will not be truly without limit. Intracellular capacity will eventually introduce a threshold to further expansion without cell bursting.

Dynamic transitions are defined quite simple. Units of $\mathbf{X}_t$ will divide and increase according to the standard mechanism seen in a birth process.

$$\Pr \{ \Delta \mathbf{X}_t = 1, \Delta \mathbf{Y}_t = 0 \} = \lambda \mathbf{X}_t \Delta t + o(\Delta t)$$

Suppressive interactions occur with a rate proportional to the product of the two counts, and result in a single elimination of each type.

$$\Pr \{ \Delta \mathbf{X}_t = -1, \Delta \mathbf{Y}_t = -1 \} = \alpha \mathbf{X}_t \mathbf{Y}_t \Delta t + o(\Delta t)$$

Hence, with just the addition of a no-event possibility,

$$\Pr \{ \Delta \mathbf{X}_t = 0, \Delta \mathbf{Y}_t = 0 \} = 1 - (\lambda \mathbf{X}_t + \alpha \mathbf{X}_t \mathbf{Y}_t) \Delta t + o(\Delta t)$$

the continuous time Markov chain is fully defined over infinitesimal durations.

Our focus of interest is in understanding the relationship between incident pathogen count and infection survivability. In other words the probability

$$\Pr \{ \text{infection survival} \} = 1 - \lim_{t \rightarrow \infty} \Pr \{ \mathbf{X}_t = 0 \},$$

signifying late time non-extinction. Cases where survival ultimately persists will include the complete exhaustion of antagonistic granules ($\mathbf{Y}_t = 0$). One may devise a model in which the granules are produced at some rate by the cell, but for now we will aim to understand this more simple version where an initial immune capacity is fixed ($\mathbf{Y}_0$) and never supplemented.

1.2 Mathematical definition

CT Markov process,

$$\mathcal{X} = \{ (\mathbf{X}_t, \mathbf{Y}_t) : t \geq 0 \},$$

state space.

$$\mathcal{S} = \{ (i, j) \mid i \in \mathbb{N} \cup \{0\}, j \in \{0, \dots, N\} \}.$$

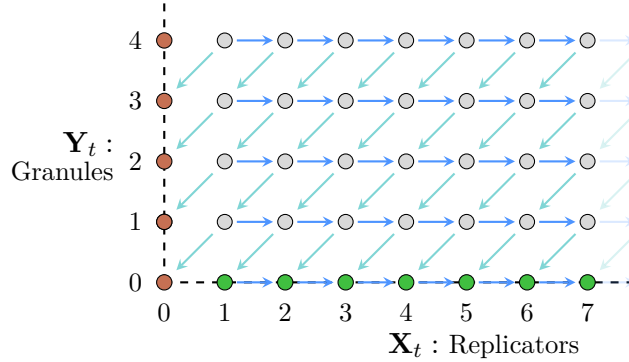
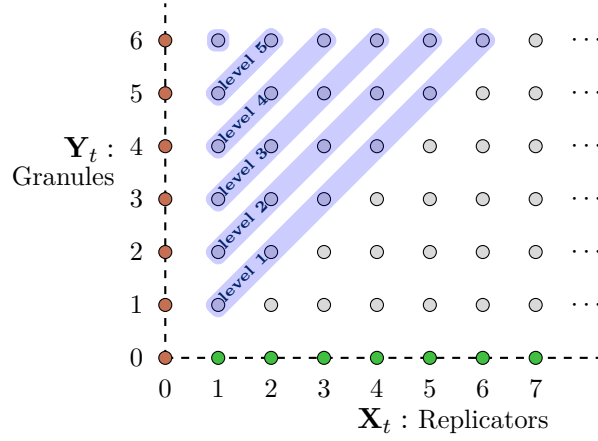
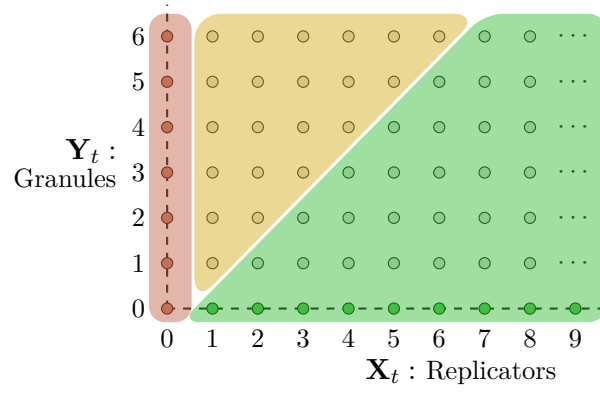
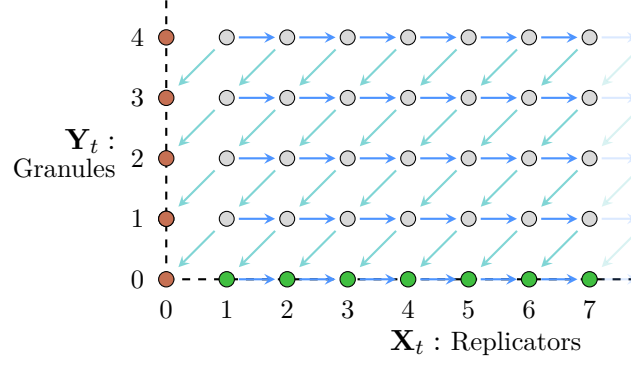


Figure 4: State space $\mathcal{S}$ with Ruptured state R in red on the right; green recovery state is seen at $(0,0)$. Arrows show possible transitions between states: blue for births and deaths of bacteria type a , those of type b are represented by orange, cyan for differentiations from type a to b , and red for the macrophage rupture event.



We can define the process by its 1 step transition probabilities,

$$p_{(i,j) \rightarrow \underline{x}}(\Delta t) = \Pr \{ \mathbf{X}_{t+\Delta t} = \underline{x} | \mathbf{X}_t = (i, j) \},$$

for states $(n_a, n_b), x \in \mathcal{S}$. Considering each of the various steps that can be taken from an arbitrary

state (n_a, n_b) these are as follows.

$$p_{(i,j) \rightarrow \underline{x}}(\Delta t) = \begin{cases} \alpha_i \Delta t + o(\Delta t) & \underline{x} = (i+1, j) \\ \beta_{i,j} \Delta t + o(\Delta t) & \underline{x} = (i-1, j-1) \\ 1 - (\alpha_i + \beta_{i,j}) \Delta t + o(\Delta t) & \underline{x} = (i, j) \end{cases}$$

(for $\Delta t \rightarrow 0$).

probability of fixation in set $(\mathbf{X}_t = 0, \mathbf{Y}_t)$

$$\theta_{i,j}$$

$$\theta_{i,j} = \hat{\alpha}_{i,j} \cdot \theta_{i+1,j} + \hat{\beta}_{i,j} \cdot \theta_{i-1,j-1}$$

$$\hat{\alpha}_{i,j} = \frac{\alpha_i}{\alpha_i + \beta_{i,j}}$$

$$\hat{\beta}_{i,j} = \frac{\beta_{i,j}}{\alpha_i + \beta_{i,j}}$$

first step analysis matrix equation, $\mathbf{p}^{(0)} = \mathbf{A} \cdot \mathbf{p}^{(0)} + \mathbf{c}$

$$\begin{bmatrix} \theta_{1,1} \\ \theta_{2,2} \\ \theta_{3,3} \\ \theta_{4,4} \\ \theta_{5,5} \\ \theta_{1,2} \\ \theta_{2,3} \\ \theta_{3,4} \\ \theta_{4,5} \\ \theta_{1,3} \\ \theta_{2,4} \\ \theta_{3,5} \\ \theta_{1,4} \\ \theta_{2,5} \\ \theta_{1,5} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hat{\beta}_{2,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \hat{\beta}_{3,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \hat{\beta}_{4,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \hat{\beta}_{5,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & \hat{\alpha}_{1,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \hat{\alpha}_{2,3} & 0 & 0 & \hat{\beta}_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \hat{\alpha}_{3,4} & 0 & 0 & \hat{\beta}_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \hat{\alpha}_{4,5} & 0 & 0 & \hat{\beta}_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & \hat{\alpha}_{1,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \hat{\alpha}_{2,4} & 0 & \hat{\beta}_{2,4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \hat{\alpha}_{3,5} & 0 & \hat{\beta}_{3,5} & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \hat{\alpha}_{1,4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \hat{\alpha}_{2,5} & \hat{\beta}_{2,5} & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \hat{\alpha}_{1,5} & 0 & 0 \end{bmatrix} \begin{bmatrix} \theta_{1,1} \\ \theta_{2,2} \\ \theta_{3,3} \\ \theta_{4,4} \\ \theta_{5,5} \\ \theta_{1,2} \\ \theta_{2,3} \\ \theta_{3,4} \\ \theta_{4,5} \\ \theta_{1,3} \\ \theta_{2,4} \\ \theta_{3,5} \\ \theta_{1,4} \\ \theta_{2,5} \\ \theta_{1,5} \end{bmatrix} + \begin{bmatrix} \hat{\beta}_{1,1} \\ 0 \\ 0 \\ 0 \\ 0 \\ \hat{\beta}_{1,2} \\ 0 \\ 0 \\ 0 \\ \hat{\beta}_{1,3} \\ 0 \\ 0 \\ \hat{\beta}_{1,4} \\ 0 \\ \hat{\beta}_{1,5} \end{bmatrix}.$$

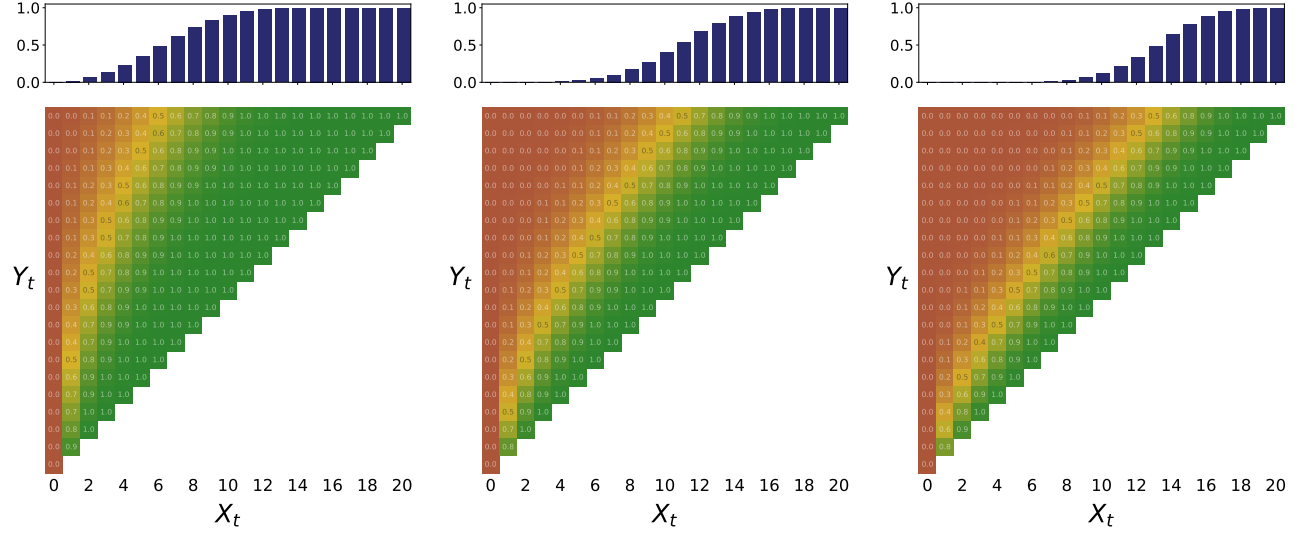


Figure 5: Probabilities of infection survival given initial count of pathogen (X_t), and granules (Y_t).

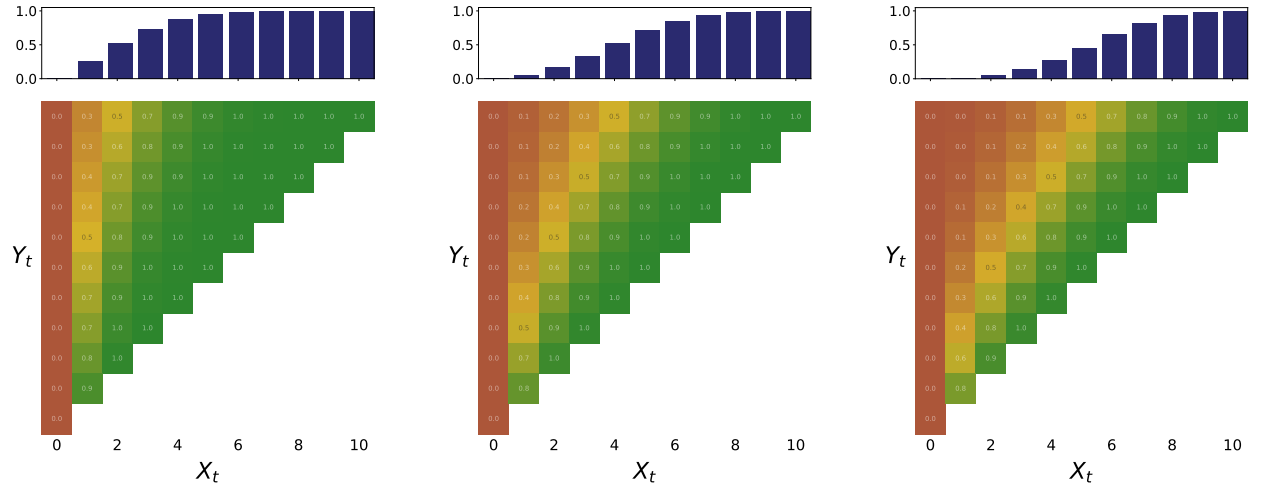


Figure 6: For each count of immune granules, there is a critical requirement of incident pathogen for infection persistence to be likely. This is all to demonstrate that immune granules create a situation in which bursting dynamics with group transfer could be advantageous. The following section addresses the same sort of dynamics but with regards to extracellular immune factors.

```
import numpy as np
```

```
N = 20
```

```
beta = 0.2
```

```
alpha = 1
```

```
def b_hat(i, j):
```

```
    return (beta * j) / (alpha + beta * j)
```

1
2
3
4
5
6
7
8
9
10

```

def a_hat(i, j):
    return alpha / (alpha + beta * j)

def generate_Asub(i, j):
    '''produces the A sub-matrices'''
    dim1 = N + 1 - i
    dim2 = N + 1 - j
    Asub = np.zeros((dim1, dim2))

    if j - i == 0:
        for k in range(N - j):
            Asub[k+1][k] = b_hat(k+2, k+1+j)

    if i - j == 1:
        for k in range(N - j):
            Asub[k][k+1] = a_hat(k+1, k+1+j)

    return Asub

def concatA(Amatrix):
    '''combines the A sub-matrices into the full A matrix'''
    w = Amatrix[0].size

    # first row
    newA = Amatrix[0][0]
    for j in range(1, w):
        newA = np.append(newA, Amatrix[0][j], axis=1)

    # other rows
    for i in range(1, w):
        segmentA = Amatrix[i][0]
        for j in range(1, w):
            segmentA = np.append(segmentA, Amatrix[i][j], axis=1)
        newA = np.append(newA, segmentA, axis=0)

    return(newA)

def full_A():
    '''uses above two functions to produce full A matrix'''
    Amat = np.zeros((N, N), dtype=object)
    for i in range(1, N+1):
        for j in range(1, N+1):
            Asub = generate_Asub(i, j)
            Amat[i-1][j-1] = Asub

    A_full = concatA(Amat)
    return(A_full)

```

```

def generate_Csub(i):
    '''produces the C sub-vectors'''
    dim = N + 1 - i
    Csub = np.zeros(dim)
    Csub[0] = b_hat(1,i)

    return Csub

def concatC(Cvector):
    '''combines the C sub-vectors into the full C vector'''
    newC = Cvector[0]
    for i in range(1, Cvector.size):
        newC = np.append(newC, Cvector[i], axis=0)

    return(newC)

def full_C():
    '''uses above two functions to produce full C vector'''
    Cvec = np.zeros(N, dtype = object)

    for i in range(1, N+1):
        Csub = generate_Csub(i)
        Cvec[i-1] = Csub

    C_full = concatC(Cvec)

    return(C_full[:, np.newaxis])

def get2D_dist(theta_vector):
    '''converts the 1D array with the theta values into its 2D grid form'''
    out = np.zeros((N+1, N+1))

    out[0] = np.array([1]*(N+1))

    idx = 0
    for c in range(N):
        for r in range(N-c):
            i = r+1
            j = r+1+c

            out[i][N-j] = round(theta_vector[idx][0], 3)

            idx += 1

    return out.T

A = full_A()
C = full_C()

```

```

# prepares A matrix for linalg solve function
Amod = np.identity(A.shape[0])-A

# gets theta vector solution
solution = np.linalg.solve(Amod,C)

# converts to 2D form
dist_2D = get2D_dist(solution)

# switches result from \xt death to survival probability
dist_2D_survival = np.ones((N+1, N+1)) - dist_2D

# extracts values relevant to initial conditions
ICvec = dist_2D_survival[0]

```

119
120
121
122
123
124
125
126
127
128
129
130
131
132
133
134

2 Fixed removal

In the pursuit of modelling multi-stage intracellular infection, we have been working on an iterated birth death catastrophe process with the assumption of complete transfer (chained ruptures). In trying to determine whether there is a fundamental reason for bursting dynamics in pathogenesis, this model is of limited efficacy. With agents replicating identically whenever they are within a host cell, it matters not how they are moved, just that they are replicating. With chained ruptures, the burst event is merely a recorded instance without tangible implication to agent spread. And if instead, the bacteria modelled were to make their way one by one to a new host, or even to multiple hosts, no change would be made to the expected growth rate of the overall population. Even if some delay were introduced for unit transfer between hosts, this would merely have the effect of adding intervals of inactivity to the time axis; broad propagation dynamics would remain identical.

It seems that if we wish to begin understanding why, or if there is a reason why, pathogens follow a bursting regime of infection, we must consider additional factors. Namely, we must take into account that the death and replication rates of individual agents are not entirely independent to the surrounding cohort. For example, various of the bodies immuno-factors are depletable. For instance, leucocytes including macrophages and neutrophils contain on the order of 10 so called granules. Small packets containing oxidising compounds with a toxic effect on invading pathogens. Given these can be used up, the effective within-host death rate of infectious agent will reduce as more and more of the incident cohort are killed. There are likely extracellular immune factors which exhibit a similar pattern of depletion.

Here is presented an extension to the “chained rupture” model. There is included a simple assumption which arises from an argument of immuno-depletion: that following a burst release event, some defined fixed count of the agents are removed. This is used as the basis for attempting to understand whether bursting pathogenesis strategies might be of fundamental advantage in relevant cases.

2.1 Successive birth death catastrophe

The concept of chained ruptures is simple. A series of BDC processes follow on from one another, the result of the last going on to found an initial condition for the next.

Starting with an single replicative unit, the first process undergoes the standard dynamics of birth and death. Then, at the time of the catastrophe event, the quantity of units present is, as it were, immediately transfers into another target cell founding the second process with its initial condition as given. This is the main assumption: that released replicative units somehow find their way, all at once, to the same host cell. Of course this is quite reduced from the biological context.

A less restricted view would be that following rupture, all of the r_1 released bacteria undergo Brownian motion and drift gradually from the burst location. The cohort, as it spreads out will encounter multiple new target cells and get taken up one by one. Albeit less so, this is also greatly reduced from biological reality, which in its full complexity will be beyond the scope of concise mathematical description. For example, the target cells we have focused on mostly are macrophages. Macrophages do not merely float around in Brownian drift; they are motile, and essentially walk around their environment seeking out debris and alien material to engulf and make away with. The agents themselves are little more straightforward. *Yersinia pestis* can “decide” whether it will be absorbed or not by such phagocytes. In addition to this, the bacteria can inject immune cells with a host of special proteins with various effects including neutralisation.

Generally, the playing field of these actors is just that, an environment of play, in which the responsive agents interact with one another; causal relationships ensue which go far beyond mere drift and chance.

2.2 Extracellular removal

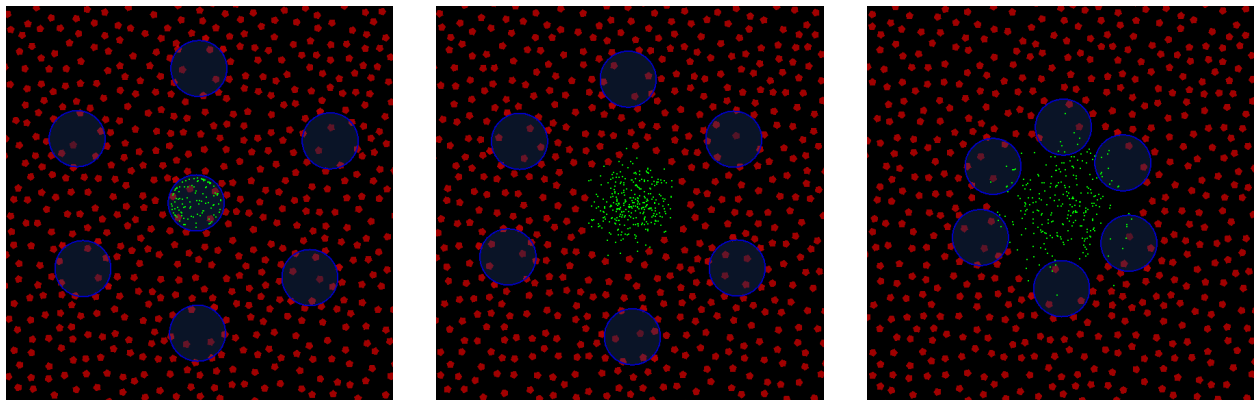


Figure 7: In the above simulation run snapshots, it is demonstrated that given the presence of depletable extracellular immuno factors, a larger released pathogenic load creates a safety in numbers "flooding effect" leading to more incident pathogen in surrounding host cells.

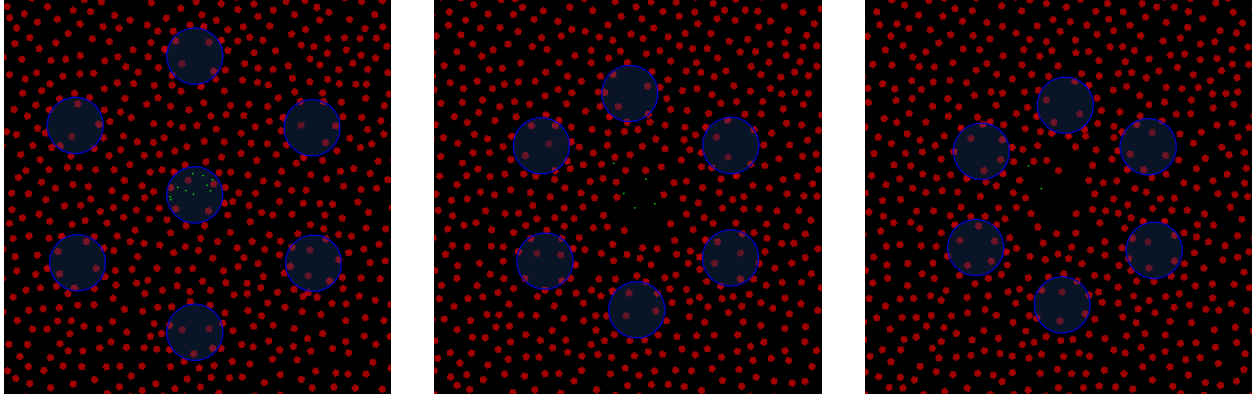


Figure 8: Contrasting the above situation, a small released load may be depleted without persistence to the next rounds of intracellular absorption and replication.

2.2.1 The assumption

But let us humour it — this concept of spatial drift. Despite imperfection, it does improve on the simpler dynamic of chained immediate transfer. Picture the playing field: a three dimensional extracellular matrix populated by three types of agents: replicative units (pathogen), Host cells (Macrophages), and another type of immuno-active cell; we’ll call them simply “killer cells”. This third type is able to eliminate the units of pathogen on contact, being used up in the process; one to one encounters where a killer cell removes a replicative agent and itself from play.

To begin with, the extracellular matrix is populated evenly with target cells and killer cells, each group occupying the space in a uniform spatial density. Killer cells are smaller and more numerous than target cells; a greater number per unit volume. Dynamics within target cells are as we understand in the birth death catastrophe process. The units of pathogen replicate and die within the cell until eventually rupture is caused and the load is released. In this model, we are considering movement to be Brownian. An extracellular colony will drift from their burst location, spreading out and coming into contact with surrounding cells. Some will encounter target cells, others, killer cells. Now... my intention with this is not to construct a full mathematical description of this model game. This would be cumbersome and intractable. It will be possible to simulate it computationally, and this may prove helpful in supporting the assumption which I hope to use the model to justify. If we suppose that no new killer cells are introduced into the environment. An assumption that pathogen movement and absorption occurs on a shorter time scale than killer generation. Then, with greater burst sizes, each individual bacteria will have a smaller probability of being removed. Those at the leading edge, as it were, clearing the way for their comrades in the rear.

If it happens that free agents do tend to find a new host before long — an assumption taken to the limit in the case of chained immediate transfer — this above observation offers the basis of an approximate mathematical rule. That is, there will be a relationship between burst size and the number of pathogen killed in the process of transfer.

As bursts of greater and greater number are compared, the casualty count will increase more and more slowly. Qualitatively this might be like a logarithmic progression.

$$\theta(r) \sim \log(r + 1),$$

where r is a given burst size and θ is the number of replicative agents removed by killer cells. (The +1 is simply to make the curve go through the origin).

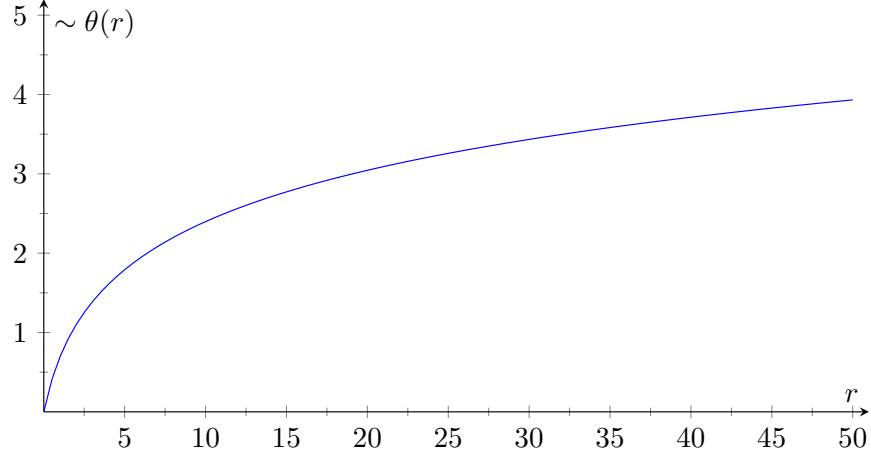


Figure 9: Qualitative representation of the diminishing removed count $\theta(r)$, given released numbers.

Given this to be the case, it might prove reasonable to impose a fixed removal following release. In this construction, the number removed by killer cells will be a specified constant, regardless of burst size. Though, if the burst size is less than this constant, simply all of the units will be killed.

$$\text{removal} = \begin{cases} r, & \text{if } r \leq \theta \\ \theta, & \text{if } r > \theta. \end{cases}$$

Now taking θ to be a constant.

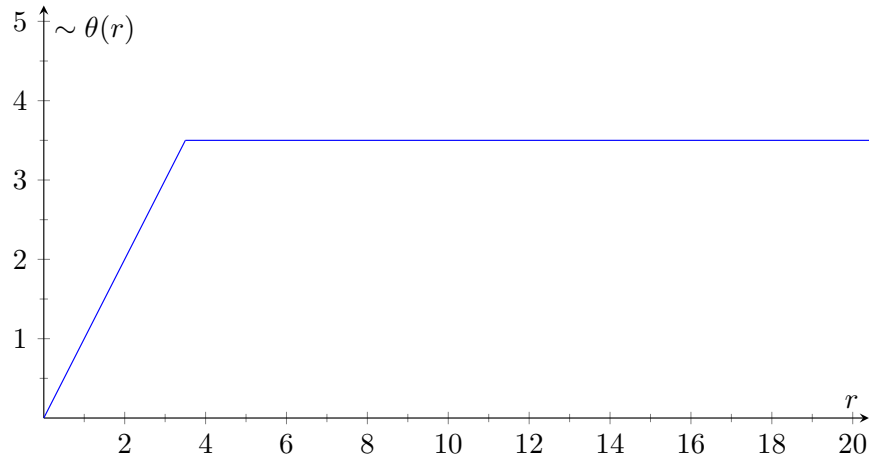


Figure 10: An approximate representation of the above shown (figure above) diminishing removal effect. Fixed removal count θ facilitates analysis of proliferation number. (the number of new infections provided by a single burst event assuming one pathogen per new host.)

The potential efficacy of this assumption rests in the likelihood that the same qualitative dynamics

will be shared with the case of logarithmic removal (or whatever the actual relation is, assuming it is of this sort).

There are a couple of points to address.

- I want to impose a further assumption: that at the time of each rupture event, the killer cell prevalence has returned to its steady state and is once again of uniform spatial distribution.
- We may frame the “fixed removal” dynamic in the context of chained transfers where all agents released have the same target cell to go to. Or, we may consider a case in which each successful pathogenic unit goes on to found an infection in a new target cell.

Aside: In the case where all surviving agents go directly to the same host cell, we can potentially frame the replication ratio as follows: Expected number of surviving released units *per incident agent*.

Let us for now take this second option — that each released cell goes on to found a new infection in a new target cell. We can consider the effective replication number between target cell infections. That is, the expected number of newly infected target cells created by a given host of one initial bacterium. Because each surviving released infectious particle infects a new host, the replication number is simply the expected quantity of these survivors.

2.2.2 Intra-burst replication ratio

To begin with, we must define r : the outcome of a BDC process with a single incident unit. Mainly, we must choose, whether or not to include outcomes that are 0 due to process death. It makes practical sense for our purpose to choose this option, to not ignore cases of $r = 0$. With this $\mathbb{E}(r) = \lim_{t \rightarrow \infty} \mathbb{E}(\mathbf{Y}_t) = V_{\text{inf}}$.

The number that survive our fixed removal will be $\max(0, r - \theta)$. Hence, the expected replication number is

$$R_0 = \mathbb{E}(\max(r - \theta, 0)).$$

We can deconstruct this based on whether the burst size is bigger or smaller than θ :

$$\begin{aligned} \mathbb{E}(\max(0, r - \theta)) &= \mathbb{E}(r - \theta | r \geq \theta) \Pr\{r \geq \theta\} + \mathbb{E}(0) \Pr\{r < \theta\} \\ &= [\mathbb{E}(r | r \geq \theta) - \theta] \Pr\{r \geq \theta\} \end{aligned}$$

Then, we can extract $\mathbb{E}(r | r \geq \theta)$ in the following way.

$$\mathbb{E}(r) = V_{\text{inf}} = \mathbb{E}(r | r \geq \theta) \Pr\{r \geq \theta\} + \mathbb{E}(r | r < \theta) \Pr\{r < \theta\}$$

With

$$\Pr\{r < \theta\} = \sum_{i=0}^{\theta-1} \Pr\{r = i\},$$

$$\Pr\{r \geq \theta\} = 1 - \Pr\{r < \theta\},$$

and,

$$\mathbb{E}(r | r < \theta) = \sum_{i=1}^{\theta-1} i \Pr\{r = i | r < \theta\} = \sum_{i=1}^{\theta-1} i \frac{\Pr\{r = i\}}{\Pr\{r < \theta\}},$$

we can make the rearrangement,

$$\begin{aligned}
\mathbb{E}(r|r \geq \theta) &= \frac{\mathbb{E}(r) - \mathbb{E}(r|r < \theta)\Pr\{r < \theta\}}{\Pr\{r \geq \theta\}} \\
&= \frac{\mathbb{E}(r) - \sum_{i=1}^{\theta-1} i \frac{\Pr\{r=i\}}{\Pr\{r < \theta\}} \Pr\{r < \theta\}}{\Pr\{r \geq \theta\}} \\
&= \frac{\mathbb{E}(r) - \sum_{i=1}^{\theta-1} i \Pr\{r = i\}}{1 - \sum_{i=0}^{\theta-1} \Pr\{r = i\}}.
\end{aligned}$$

This is analytically computable because we have

$$\begin{aligned}
\Pr\{r = i\} &= \frac{\delta}{\lambda} \left(\frac{1}{a}\right)^i \\
\mathbb{E}(r) &= V_{\text{inf}}.
\end{aligned}$$

Hence,

$$\mathbb{E}(\max(0, r - \theta)) = \left[\frac{\mathbb{E}(r) - \sum_{i=1}^{\theta-1} i \Pr\{r = i\}}{1 - \sum_{i=0}^{\theta-1} \Pr\{r = i\}} - \theta \right] \Pr\{r \geq \theta\}$$

becoming

$$\mathbb{E}(\max(0, r - \theta)) = \left[\frac{V_{\text{inf}} - \mathcal{K}}{1 - \mathcal{G}} - \theta \right] (1 - \mathcal{G}) = V_{\text{inf}} - \mathcal{K} - \theta(1 - \mathcal{G})$$

where

$$\mathcal{G} = b + \sum_{i=1}^{\theta-1} \frac{\delta}{\lambda} \left(\frac{1}{a}\right)^i \quad \text{and} \quad \mathcal{K} = \sum_{i=1}^{\theta-1} i \frac{\delta}{\lambda} \left(\frac{1}{a}\right)^i.$$

Because the formula

$$\Pr\{r = i\} = \frac{\delta}{\lambda} \left(\frac{1}{a}\right)^i$$

does not correctly apply when $i = 0$, this is why the sum for $\mathcal{G}$ begins at 1, and why there is the term, b , preceding the sum. This is the probability that the process ultimately ends in death, an output of 0.

Direction. The primary interest is in whether this value is greater or less than 1, and if greater, by how much. And the question is in how this changes with δ . Do larger burst sizes make an infection more viable and proliferative?

Before proceeding with this analysis, we should verify whether the formula is correct using numerical simulation comparisons.

2.2.3 Verification of formula

Compare replication ratio from formula with equivalent numerical results from simulations ...

We must be clear as to whether r includes bust cases alone, or whether it factors in cases of death. Bear in mind that the formula for $\Pr\{r = i\}$ is not defined when $i = 0$, in this case, $\Pr\{r = 0\} = I_-$.

An observation: The formula for $\Pr\{r = i\}$, does not apply to $i = 0$. We know the probability that $r = 0$. It's I_- , or b , or $\eta^2 + \sqrt{\eta^2 - \mu/\lambda}$. Furthermore, it can be shown that

$$I_- + \sum_{i=1}^{\infty} \Pr\{r = i\} = 1 :$$

$$\sum_{i=1}^{\infty} \frac{\delta}{\lambda} \left(\frac{1}{a}\right)^i = 1 - b.$$

Then using the geometric sum formula $\sum_{k=1}^{\infty} y^k = y/(1 - y)$ ($|y| < 1$),

$$\frac{\delta}{\lambda} \left(\frac{\frac{1}{a}}{1 - \frac{1}{a}}\right) = 1 - b$$

$$\frac{\delta}{\lambda} = (1 - b)(a - 1) = (a + b) - ab - 1$$

With $a + b = \frac{\lambda + \mu + \delta}{\lambda}$ and $ab = \frac{\mu}{\lambda}$, it can be shown that the LHS does equal the RHS,

$$\frac{\delta}{\lambda} = \frac{\lambda + \mu + \delta}{\lambda} - \frac{\mu}{\lambda} - 1 = \frac{\delta}{\lambda}$$

. This is all to say that we can be quite certain of the fact that

$$\mathcal{G} = \Pr\{r < \theta\} = I_- + \sum_{i=0}^{\theta-1} \frac{\delta}{\lambda} \left(\frac{1}{a}\right)^i$$

2.2.4 Analysis of proliferability

Basically, how does the replication ratio change depending on the burst rate (burst size) ...

To begin with, we can take the formula found,

$$\mathbb{E}(\max(0, r - \theta)) = V_{\inf} - \mathcal{K} - \theta(1 - \mathcal{G}),$$

and make it easier to analyse. Because $|\frac{1}{a}| < 1$, we can use the following geometric sum identities to re-write $\mathcal{G}$ and $\mathcal{K}$.

$$\begin{aligned} \sum_{i=1}^{n-1} &= \frac{s(1 - s^{n-1})}{1 - s} \\ \sum_{i=1}^{n-1} &= \frac{s(1 - ns^{n-1} + (n-1)s^n)}{(1 - s)^2}. \end{aligned}$$

With, $s = \frac{1}{a}$ and $n = \theta$, it can be shown that $\mathbb{E}(\max(0, r - \theta))$ becomes

$$\mathbb{E}(\max(0, r - \theta)) = V_{\infty} - \theta(1 - b) + \frac{\delta}{\lambda} \left(\frac{\frac{1}{a} \left(\theta - 1 - \theta \frac{1}{a} + \left(\frac{1}{a}\right)^{\theta} \right)}{\left(1 - \frac{1}{a}\right)^2} \right)$$

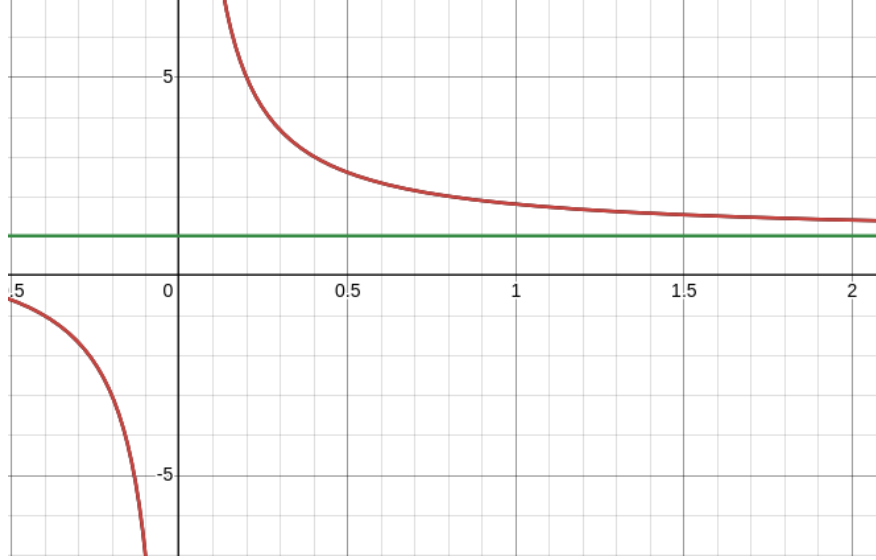


Figure 11: $\lambda = 0.8$ $\theta = 0$. x-axis: δ . With smaller delta, there is an increasingly large burst size. In this plot the removal value, θ is 0. On the y-axis is the quantity $\mathbb{E}(\max(0, r - \theta))$ and $\mathbb{E}(r)$, the mean release without removal. When $\theta = 0$, these two values are obviously the same, and the lines overlap. In the figures below, $\theta \neq 0$, and the lines do not overlap.

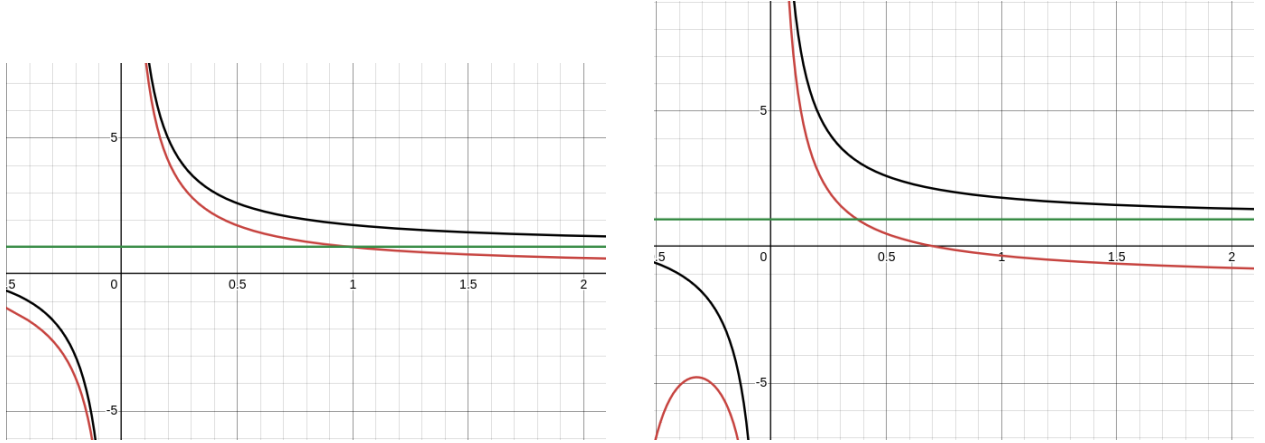


Figure 12: It can be seen that when $\theta > 0$, the required δ to ensure supercritical proliferation ($\mathbb{E}(\max(0, r - \theta)) > 1$) is smaller (the required release size is higher). ($\lambda = 0.8$, $\theta = 0.8$ and $\theta = 2.3$ (left and right respectively))

2.2.5 Motivation

I am interested in the question of whether there is some fundamental advantage to bursting dynamics in pathogenesis. In the case of *F. tularensis* and *Y. pestis*, host cell lysis may be caused merely as the simplest means of release. With bacteria being relatively large in comparison to the protein complexes that constitute a cell membrane, it would be difficult to employ a strategy in which some but not all of the pathogen are released. The case of viruses is different. They are sufficiently small to allow for partial release of an intracellular load. To be fair, in the case of viruses such as HIV, there is an explicit advantage to not killing the cell: its machinery continues to produce virions.

Despite this, I have a suspicion that there is some advantage to bursting style strategies. That is, where agents are released in large groups all at once from their hidden position within host cells. As opposed to continually, or in dribs and drabs. There are a great many stories one could tell about the reason for which this might be the case, that there is an advantage to bursting. Here are three categories in which the ones I've thought up fall under.

- **Subductive proliferation.** Replication rate reduces over time within a host cell, perhaps following recognition, or by the using up of available resources. (Though to be honest, I'm starting to think that this doesn't check out as a reason — some analysis should reveal the truth one way or the other).
- **Limited extracellular killing.** As described above, the means of removing agents are finite on relevant timescales, this leading to an effective reduction in death rate for invading pathogen. Note: this dynamic may take place either inside the cell or extracellularly as described above. For the case of within cell dynamics, some host cells contain oxidative granules, which in their finite number, destroy intracellular agents.
- **Detection avoidance** For the case of HIV, where observations have been made that release is sporadic and clustered. There may be some argument against a strategy of continual budding because it might lead to more effective detection by targeting immune cells which induce apoptosis.

Each argument of these three types essentially boils down to the catching off guard of some immune mechanism. As it were, surprising the systems of regulation to the extent that response is too slow.

2.2.6 Potential for *Y. pestis*

We might be able to frame this model in the context of *Y. pestis* and neutrophils. That is, the target cells are macrophages, the killer cells are neutrophils. And the reason for the two phases of infection is argued for by firstly, the advantage of bursting in the case of constant density killer cells, then secondly, the mass increase in killer cells causes a need to become extracellular. But this would beg the question: why not just replicate extracellularly from the beginning. There must be some advantage to being within macrophages. For one thing, they are a means of transport. Taking the agents to lymph nodes where there is lots of food. They may provide food themselves.

3 Note about quad tree sampling

To speed up the animations, I ported an open-source implementation of quadtree sampling from JavaScript to C++. It works by sorting all of the particles into a structured index (characterized by hierarchical branching of squares within squares). When checking if one of the particles intersects with the macrophage's circle, only those local to the macrophage are checked. Locality is determined and identified by the position in the quadtree. Below, there is a figure in which the local regions of the macrophage are shown in pink. Only the green dots in those squares are checked for collisions. The entire tree index needs to be recreated every frame, but despite this, the speed increase is worth it, especially in the case shown in the image on the right.

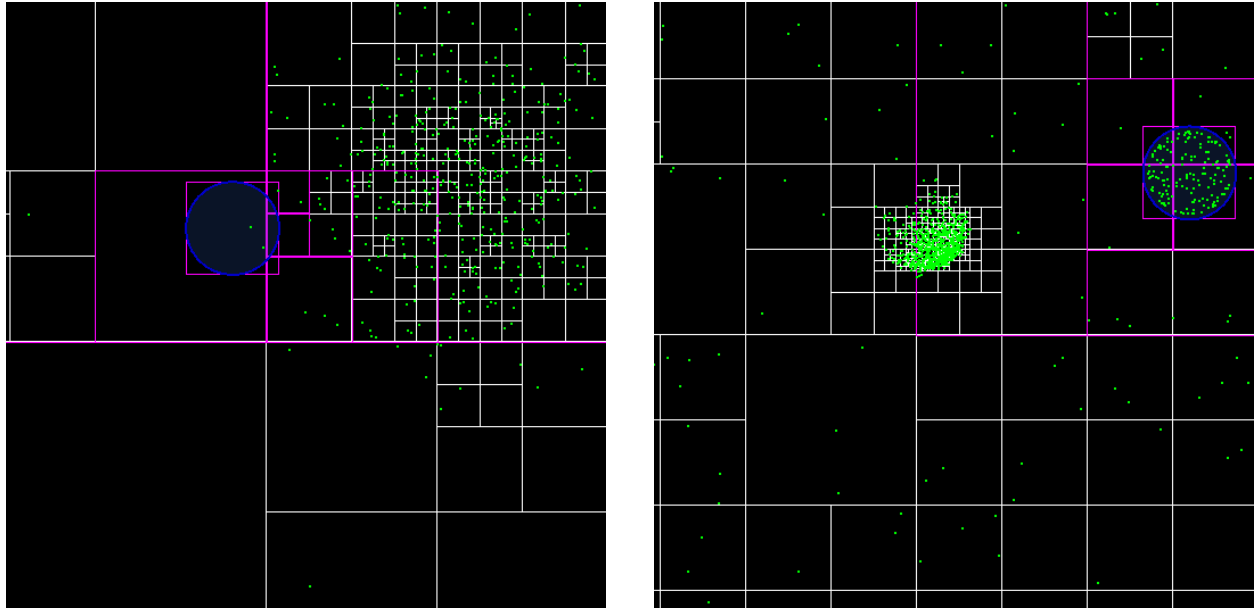


Figure 13: Visualization of the quadtree sampling used in the particle simulations shown above. Please refer to Wikipedia for more information on this; it’s quite a well-worn topic in computer graphics.

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